Total No. of Questions : 6]

SEAT No. :

P4953

[Total No. of Pages :3

BE/In Sem. - 19
B.E. (Mechanical)
RELIABILITY ENGINEERING
(2012 Course) (Elective - I) (Semester - I)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***All questions are compulsory i.e. Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.***
2) ***Figures to the right indicate full marks.***
3) ***Use of logarithmic tables slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.***
4) ***Assume suitable data, if necessary.***
5) ***Neat diagrams must be drawn wherever necessary.***

Q1) a) Define failure density and hazard rate. **[4]**

b) A series of tests were conducted to determine the strength of ten similar structures which were subjected to continuous vibratory load. The duration of survival of each structure observed is as follows: **[6]**

Specimen Number	1	2	3	4	5	6	7	8	9	10
Hours of survival	54	62	61	60	55	50	58	55	62	59

Calculate the mean time to failure (MTTF) and also obtain the mean failure rate for 62 hours of survival.

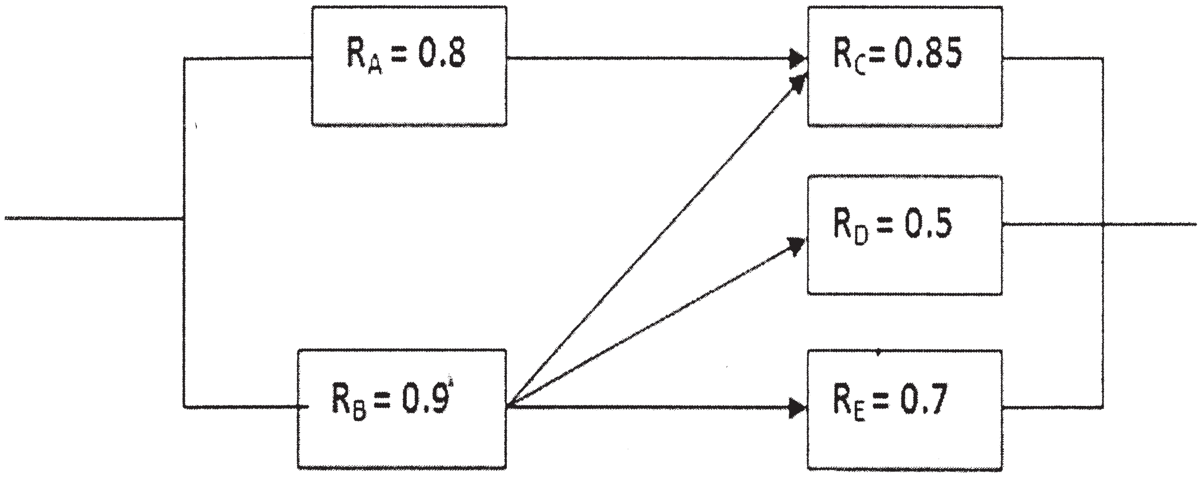
OR

Q2) a) Draw a specimen 'Bath Tub Curve' and show various regions. **[4]**

b) Compute the failure density and survival probability for the following data: **[6]**

Time interval (hrs)	0-1000	1001-2000	2001-3000	3001-4000	4001-5000	5001-6000
Failure in the interval	59	24	29	30	17	13

Q3) a) Using conditional probability method, evaluate reliability of the system shown in the following Figure: (Consider 'C' as the critical component). **[8]**

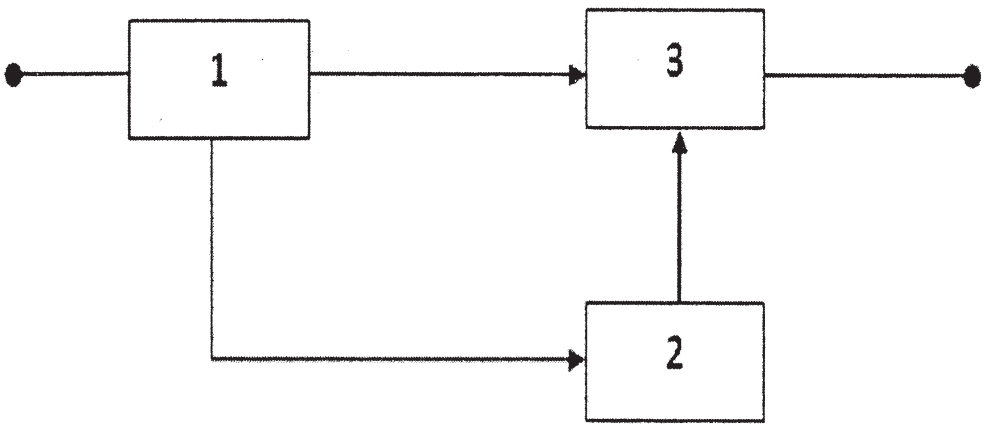


b) A system consists of four identical subsystems in parallel. What should be the reliability of each subsystem if the system reliability is to be equal to 0.97. **[2]**

OR

Q4) a) Name any four probability distributions & their selection criterion. **[4]**

b) For the block diagram shown in Figure below, write down the minimal tie-sets and cut-sets. From these, obtain the reliability of the system assuming the elements to be independent by both the tie-set and cut-set methods. **[6]**



Q5) a) Four units are connected in series with reliabilities 0.8, 0.95, 0.85 and 0.9. Calculate the system reliability. If reliability is to be increased to value of 0.65, how should this be apportioned among the four units according to the minimum effort method? **[8]**

b) What is reliability apportionment? List all the apportionment techniques. **[2]**

OR

Q6) A system is composed of four units connected in series. The failure rates for these units are $\lambda_1 = 0.003, \lambda_2 = 0.007, \lambda_3 = 0.004 \text{ and } \lambda_4 = 0.002$. A system requires a reliability of 0.9 for 10 hours of operation. Allocate reliabilities to the four units. **[10]**

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